

AD7190

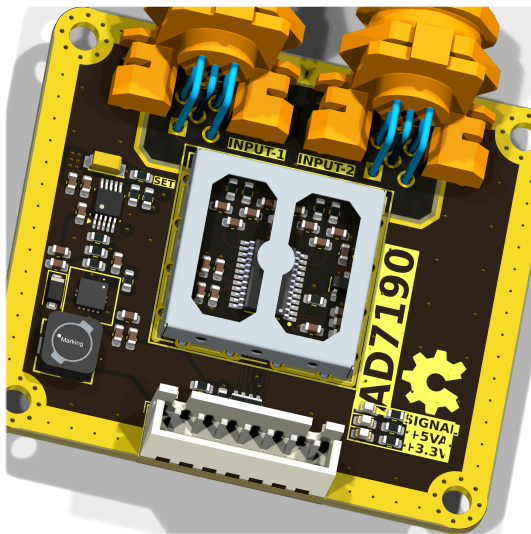
2026-10-08

Variant:

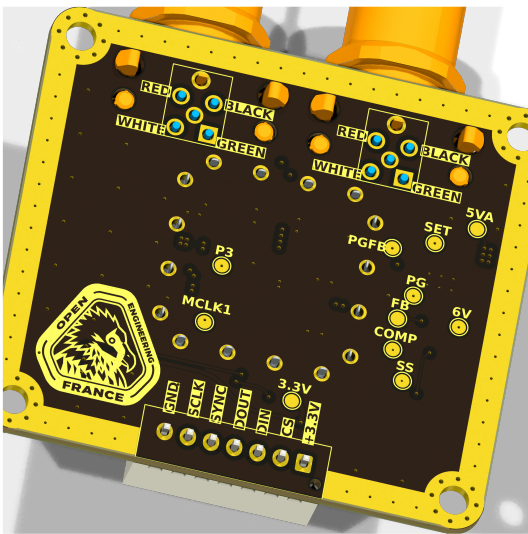
Rev + (Unreleased)

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TOP VIEW



BOTTOM VIEW



NOTES

OPEN\_WEIGHT - AD7190 24-bit load-cell front-end, part of OPEN THRUST LAB.  
Single +3.3 V host supply, SPI interface, 2 x 4-wire bridge inputs.

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 08-Oct-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:

+3.3V host -> TPS61086  
6 V -> LT3042 +5VA (AVDD + excitation)

DEBUG NOTE:

TP on every rail.  
Wait >= 600 ms (PG) before ADC init.

DESIGN NOTE:

SPI mode 3,  
SCLK <= 5 MHz,  
RDY on DOUT.

DESIGN NOTE:

REFIN1 = bridge excitation: ratiometric.

LAYOUT NOTE:

AFE under the WE-SHC shield H5, boost SW node away.

|                  |                                     |                              |                     |                              |                      |
|------------------|-------------------------------------|------------------------------|---------------------|------------------------------|----------------------|
|                  | Comments:                           | Company:<br>OPEN THRUST LAB  |                     | Variant:                     | Git Hash:<br>b3867f9 |
|                  | Title:<br>AD7190                    | Board Name:<br>AD7190        |                     | Project Name:<br>OPEN_WEIGHT |                      |
|                  | File Name:<br>OPEN_WEIGHT.kicad_sch | Designer:<br>SIEBERT DIMITRY | Date:<br>2026-09-29 | Revision:<br>+ (Unreleased)  |                      |
| Sheet Path:<br>/ |                                     | Reviewer:                    | Size:<br>A3         | Sheet:<br>1 of 10            |                      |